Science and Technology Facilities Council

Scientific Computing

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A Scalable Probabilistic Framework for Risk Assessment of Complex Infrastructure Networks

1. Introduction

Infrastructure networks, including transportation, power, and water systems, underpin daily life. Yet, they are increasingly exposed to natural hazards that can cause nation-wide disruption. The complexity of these networks, coupled with the uncertainty of the threats they face, makes them challenging to assess. Addressing this challenge requires a systems-thinking approach that computes the network-level consequences, rather than assessing components individually. To this end, we have developed an end-to-end risk assessment tool that addresses network complexity and supports decision-making.

2. Research Aims

Assessing network risk requires linking hazard models, components damage and system performance simulation. This task remains challenging because it requires quantifying **uncertainty** and handling the large number of **components**.
This research aims to provide an end-to-end infrastructure risk-assessment process of complex network systems, from hazard occurrence to system-level consequence:

1. **Efficient risk assessment:** scalable network risk assessment propagating hazard through asset fragility to system-level performance, with flexible network, hazard, and fragility specification
2. **Decision-oriented outputs:** system-level risk metrics supporting risk-informed decisions
3. **Lower technical barrier:** delivered via the DAFNI platform, facilitating the workflow's implementation

3. Proposed Methods

3.1 Bayesian Network

A **Bayesian network** (BN) enables such modelling naturally through its visualisation: it represents variables as nodes and their causal relationship through directed edges connecting the nodes (e.g., hazard intensity → component damage). Uncertainty can be modelled by the **probability distribution** defined for each node **conditioned** on their parent nodes.

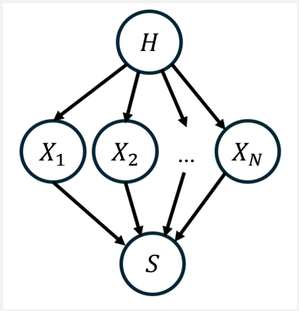


Figure 1. Illustrative BN graph structure for a system event S that comprises N components, whose states are represented by random variables X_1 ,…, X_N . These components are affected by hazard node H.

3.2 Tensor-based Bayesian Network

Our approach: Tensor-based Bayesian network (TBN) is an advanced Bayesian network modelling method, using tensor operations for efficient modelling and inference especially. TBN is especially useful for handling large-scale systems, as it can handle high-dimensional probabilistic models.

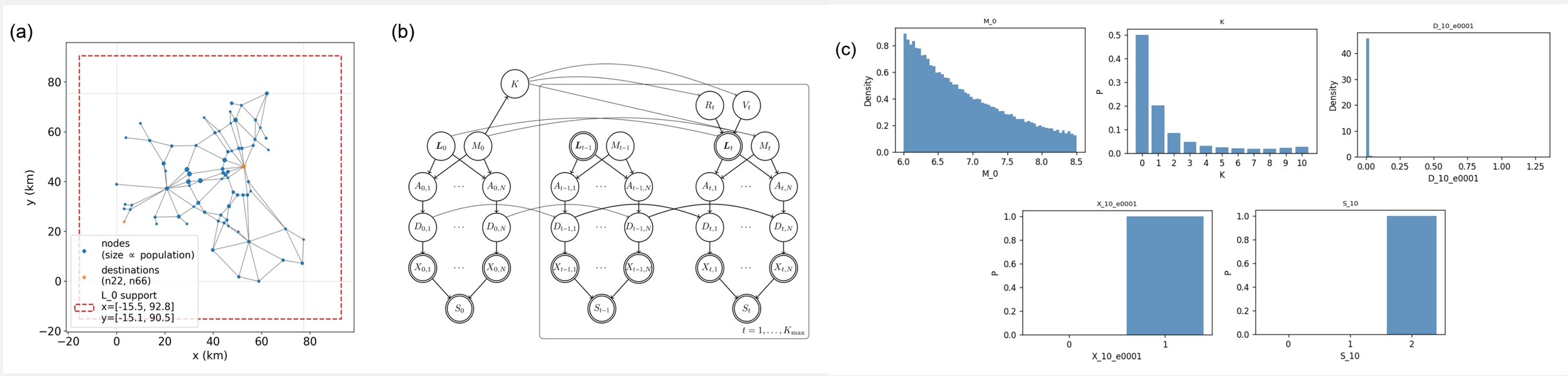


Figure 2. Overview of the EMA aftershock case study. **(a)** EMA highway network and hypothetical seismic source zone. **(b)** Bayesian network linking hazard, edge-level damage, and system-state variables. **(c)** Representative prior distributions of hazard, damage, and system-state variables from forward sampling.

4. TBN workflow on the DAFNI platform

Model: TBNpy (Python toolkit for TBN)
Data: User-defined network topology, hazard, and fragility inputs
Outputs: Probabilistic analysis summary files
Visualisation: Visualisation of system risks and geographically mapped results

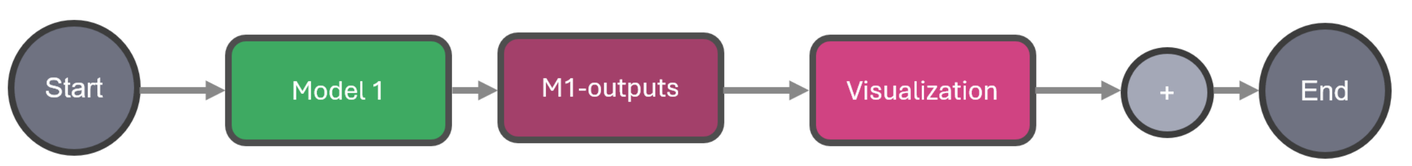


Figure 3. TBN workflow on the DAFNI platform

5. Planned Training Workshop On System Risk Assessment (expected in February 2027)

5.1 Intended Learning Outcomes

- Build and apply a tensor-based Bayesian network workflow for risk assessment of network systems
- Adapt the workflow to new infrastructure systems and hazard models

5.2 Programme structure

1. Define network topology (Fig. 2a) and hazard model (e.g., seismic, flood, etc.)
2. Build Bayesian network linking hazard, fragility, and system-state variables (Fig. 2b)
3. Run forward inference to estimate system-level performance (Fig. 3c)
4. Extract decision-oriented risk metrics for risk-informed resource prioritisation

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